

Transformers

X26 (2026) — English translation

A transformer is an electromagnetic device having two or more inductively coupled windings. It is designed to convert an AC system with one parameters into an AC system with other parameters. Most often, voltage and current are transformed, but the signal frequency does not change. In all points of the problem, you can use μ_0 to express formulaic answers where necessary. Use $\mu_0 = 4\pi \cdot 10^{-7} \text{ ГН/М}$ for calculations. Unit vectors along the directions are denoted by the letter $\mathbf{e}$. For example, the unit vector along the x axis is denoted $\hat{x}$. Give all numerical answers in SI units.

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Part A. Single-phase transformer (3.6 points)

The simplest transformer consists of a ferromagnetic core and two windings wound on it with the number of turns N_1 and N_2 . The ohmic resistance of the windings everywhere except part C can be neglected. The magnetic permeability of the core is $\mu \gg 1$, due to which complete magnetic coupling is ensured between the windings of the transformer - that is, we can assume that the magnetic field flux Φ inside the core is not dissipated.

Notes

For the positive direction of the magnetic flux in the winding, take the direction of the flux that would be created by the current flowing in this winding along the current directions indicated in the figure. For the positive direction of the magnetic flux through the cross section of the core, take the direction of the flux, indicated in the figure by the letter Φ . The positive direction of the current is indicated in the figure. The voltage on the windings is equal to the difference between the potential of the upper point and the potential of the lower point in the figure. The mutual induction coefficient of windings i and j is defined as

$$M = \frac{\partial \Phi_j}{\partial I_i},$$

where Φ_j is the flux through winding j , and I_i is the current in the winding i .

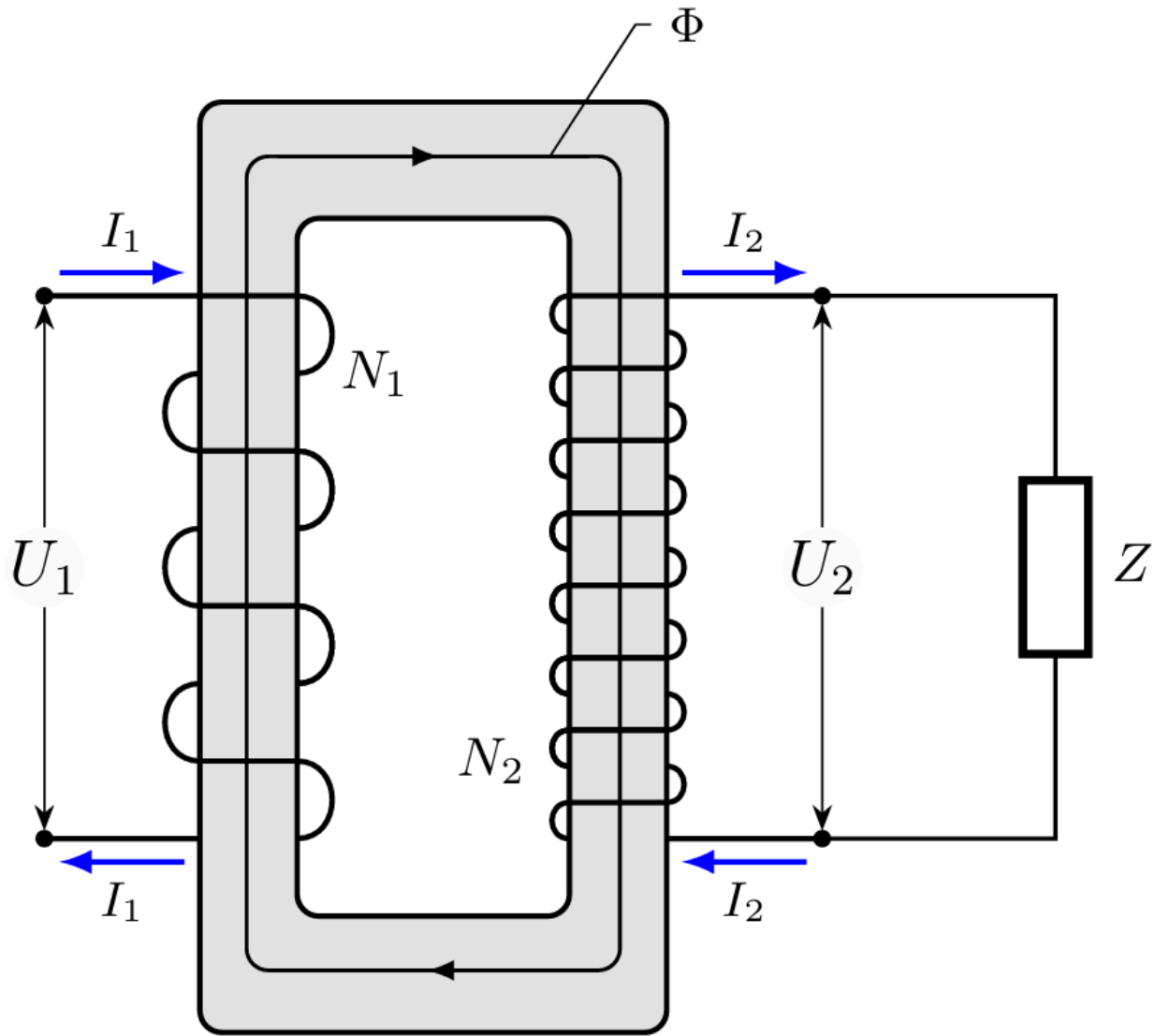


Fig. 1.

In points A1 - A3 we find the inductances of the windings L_1 , L_2 and their mutual induction coefficient M .

- A1** What are the magnetic field fluxes through the primary and secondary windings Φ_1 and Φ_2 ? **0.20**
Express your answers in terms of Φ , N_1 and N_2 .

To simplify calculations, assume that the cross-sectional area of the core S is constant. The length of the core is $\ell \gg \sqrt{S}$.

- A2** Using the circulation theorem for the magnetic field strength vector $\vec{H}$, find the magnetic field flux Φ in the core. Express your answer in terms of S , ℓ , μ , I_1 , I_2 , N_1 and N_2 . **0.20**

- A3** Obtain expressions for the winding inductances L_1 , L_2 and their mutual inductance coefficient M . Express your answer in terms of S , ℓ , μ , I_1 , I_2 , N_1 and N_2 . **0.30**

The interaction of the two windings is completely described by the found quantities L_1 , L_2 and M . In the diagrams, the transformer is depicted as follows:

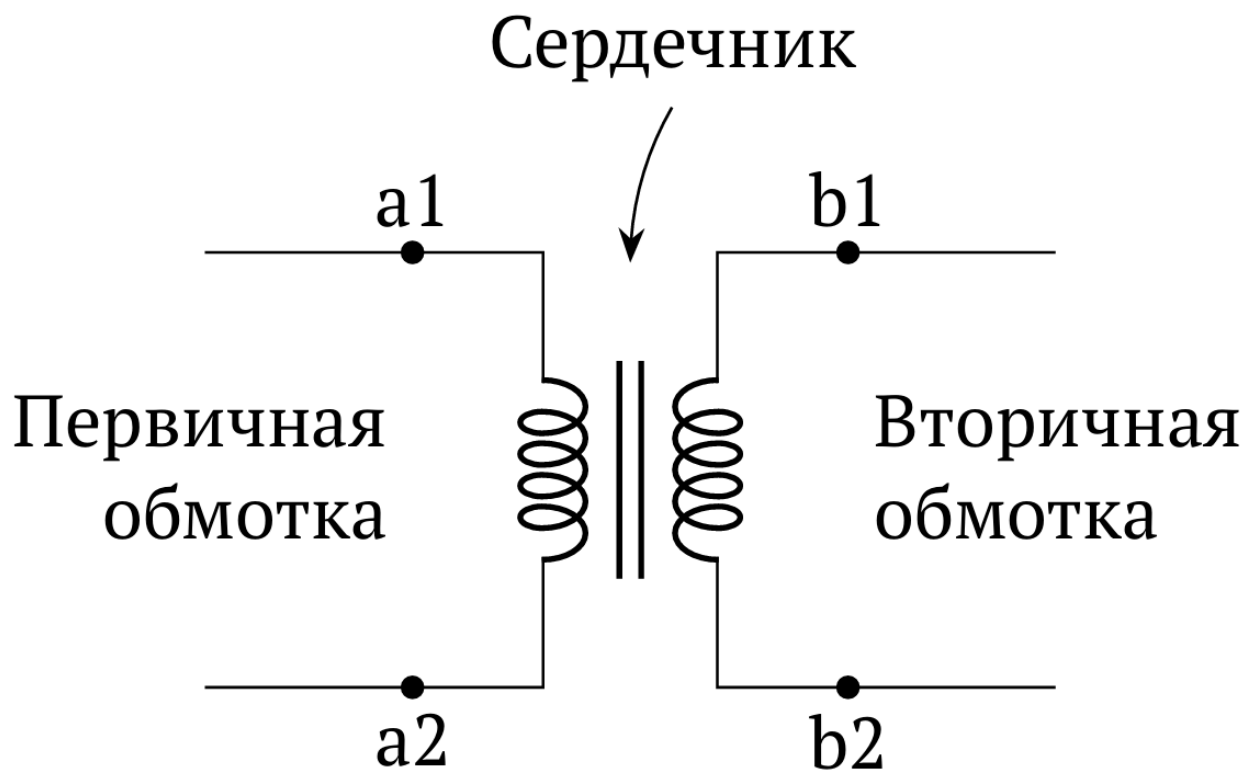


Fig. 2.

- A4** Find the magnetic induction emf $\mathcal{E}_1$ and $\mathcal{E}_2$ arising in the primary and secondary windings, respectively. Express your answer in terms of $\dot{I}_1, \dot{I}_2, L_1, L_2$ and M . Notes: EMF $\mathcal{E}_1$ is considered positive if the electrostatic potential at the top point a1 is greater than at the bottom a2. EMF $\mathcal{E}_2$ is considered positive if the electrostatic potential at the top point b1 is greater than at the bottom b2. **0.40**

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A sinusoidal voltage $U_1(t) = \text{Re} [\tilde{U}_1 e^{-i\omega t}]$ with an amplitude $U_0 = |\tilde{U}_1|$ was applied to the primary winding. All quantities in the system will also change sinusoidally, which means you can enter the corresponding complex amplitudes:

$$I_1(t) = \text{Re} [\tilde{I}_1 e^{-i\omega t}], \quad I_2(t) = \text{Re} [\tilde{I}_2 e^{-i\omega t}], \quad U_2(t) = \text{Re} [\tilde{U}_2 e^{-i\omega t}].$$

Пусть ко вторичной обмотке подключена цепь с импедансом Z . Для записи ответов вы можете явно использовать вещественную и мнимую части Z : $\text{Re } Z$ и $\text{Im } Z$ respectively.

- A5** Obtain an expression for the current transformation ratio $k_I \equiv |\tilde{I}_1|/|\tilde{I}_2|$. The response may include L_1, L_2, ω and Z . **0.50**

- A6** What is the voltage transformation ratio $k_U = |\tilde{U}_2|/|\tilde{U}_1|$? The response may include L_1, L_2, ω and Z . **0.50**

The main application of transformers is to reduce the amplitude of the current (and therefore increase the amplitude of the voltage) before applying it to the power line. This is necessary in order to reduce ohmic losses in the wires. For such transformers, called power ones:

$$k_I, k_U \gg 1 \quad (1)$$

A7 Write down the condition for the number of turns in windings N_1 and N_2 , under which the expression (1) is satisfied. **0.30**

Due to the presence of mutual induction, the voltage source U_1 can do work despite the absence of active resistance connected directly to it.

A8 Determine the phase difference $\Delta\varphi$ by which the voltage U_1 leads the current I_1 . The answer may include L_1 , L_2 , ω and Z . **0.60**

A9 Determine the time-averaged power P of source U_1 . The answer may include U_0 , Z , N_1 and N_2 . **0.60**

Part B. Instrument current transformer (0.5 points)

If you've ever done an experiment with electricity, you should know how easily an ammeter can burn out at high currents! To prevent such incidents, a measuring current transformer is used, which reduces the current by a fixed number of times. The primary winding of the transformer is a conductor with the measured alternating current I , and an ammeter is connected to the secondary winding with the number of turns N . The internal resistance of the ammeter is very small, so in fact the secondary winding of the transformer operates in a mode close to a short circuit. The magnetic permeability of the core is $\mu \gg 1$, its radius is R and is much greater than the thickness of the wire.

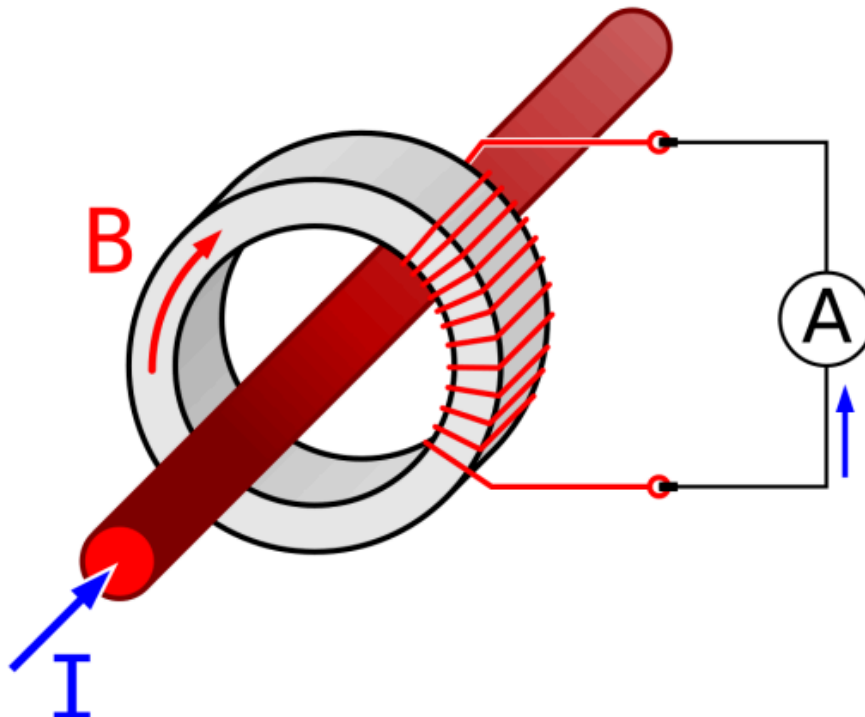


Fig. 3.

B1 Find the relation I/I_A . **0.50**

Energy losses in transformers

To use transformers for industrial purposes, it is necessary to be able to determine and minimize energy losses in them. Energy in transformers is lost mainly due to the following factors: Joule losses in the windings Joule losses in the thickness of the core due to induced eddy currents Magnetization reversal (ferromagnetic hysteresis) In the following parts, the influence of each of the factors will be studied, considering these factors to be independent and small, and the efficiency of the transformer will also be calculated. For simplicity, we will assume that the number of turns in the primary and secondary windings is the same. For numerical calculations, use the following values: Core length $\ell = 25 \text{ cm}$ Cross-sectional area of the core $S = 100 \text{ cm}^2$, its radius can be considered constant Saturation induction of the core material $B_s = 1.5 \text{ T}$ Coercivity of the core material $H_c = 20 \text{ A/m}$ Magnetic permeability of the core material $\mu = 1000$ Number of turns in each winding transformer $N = 100$ Wire diameter $d = 0.6 \text{ mm}$ Copper conductivity $\sigma_w = 5.8 \cdot 10^7 \text{ OM}^{-1} \cdot \text{m}^{-1}$ Core material conductivity $\sigma_c = 30 \cdot 10^3 \text{ OM}^{-1} \cdot \text{m}^{-1}$ Frequency of supplied voltage $f = 50 \text{ Hz}$ Thickness of magnetic core plates $h = 0.4 \text{ mm}$ Supply load resistance $R = 160 \text{ OM}$ Effective (rms) source voltage $U_{\text{rms}} = 220 \text{ V}$

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Part C. Ohmic losses in windings (2 points)

We will simulate the resistance of the windings by adding a small resistance r to the primary and secondary windings (see figure below).

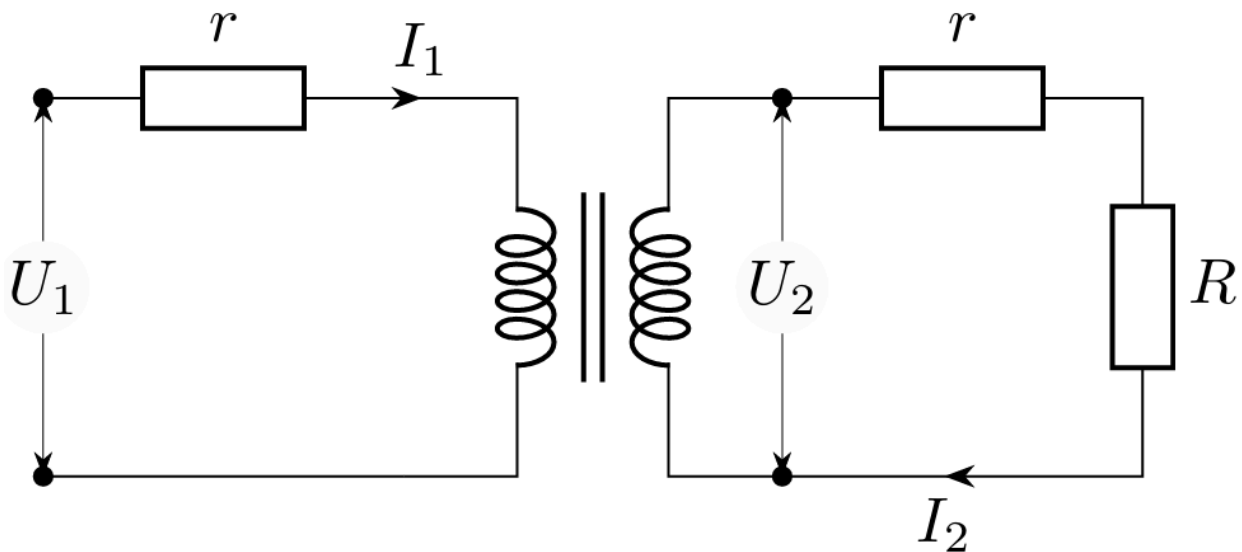


Fig. 4.

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|-----------|--|-------------|
| C1 | Obtain an expression for the average power P_0 , which is released at the load if the winding resistance can be neglected. Express your answer in terms of the amplitude of the input voltage U_0 and the load resistance R . Calculate it for a transformer whose parameters were given in the problem earlier. | 0.20 |
|-----------|--|-------------|

C2 Similar to Part A, write equations relating the complex amplitudes $\tilde{U}_1$, $\tilde{U}_2$, $\tilde{I}_1$ and $\tilde{I}_2$. The response may also include R , r , single winding inductance L and cyclic frequency ω . **0.30**

C3 Express $\tilde{I}_1$ and $\tilde{I}_2$ in terms of $\tilde{U}_1$ and other quantities you need, which may include R , r , L and ω . **0.40**

C4 Hereinafter, work in the limit $r \ll R, \omega L$. Obtain expressions for the power of Joule losses P_{r1} and P_{r2} released in the primary and secondary windings. Express your answer in terms of the amplitude of the input voltage U_0 , as well as r , ω and L . Calculate the total power of Joule losses on the windings $P_r = P_{r1} + P_{r2}$ for the transformer, the parameters of which were given earlier in the problem. Provide detailed calculations for your calculations in the solution sheets. **1.10**

Obtain expressions for the power of Joule losses P_{r1} and P_{r2} released in the primary and secondary windings. Express your answer in terms of the amplitude of the input voltage U_0 , as well as r , ω and L .

Calculate the total power of Joule losses on the windings $P_r = P_{r1} + P_{r2}$ for a transformer whose parameters are given earlier in the problem. Provide detailed calculations for your calculations in the solution sheets.

Part D. Toki Fuko (2.7 points)

If an alternating magnetic field is created in a conducting material, an eddy electric field and associated Foucault currents arise in it. This leads to additional energy losses. Let us consider a long cylindrical core with cross-section S and length ℓ (we will assume that $\ell \gg \sqrt{S}$). The magnetic field $\vec{B}(t) = \text{Re} [B_0 \hat{z} e^{-i\omega t}]$ penetrates this core along its axis z . Neglect the influence of Foucault currents on the magnetic field distribution.

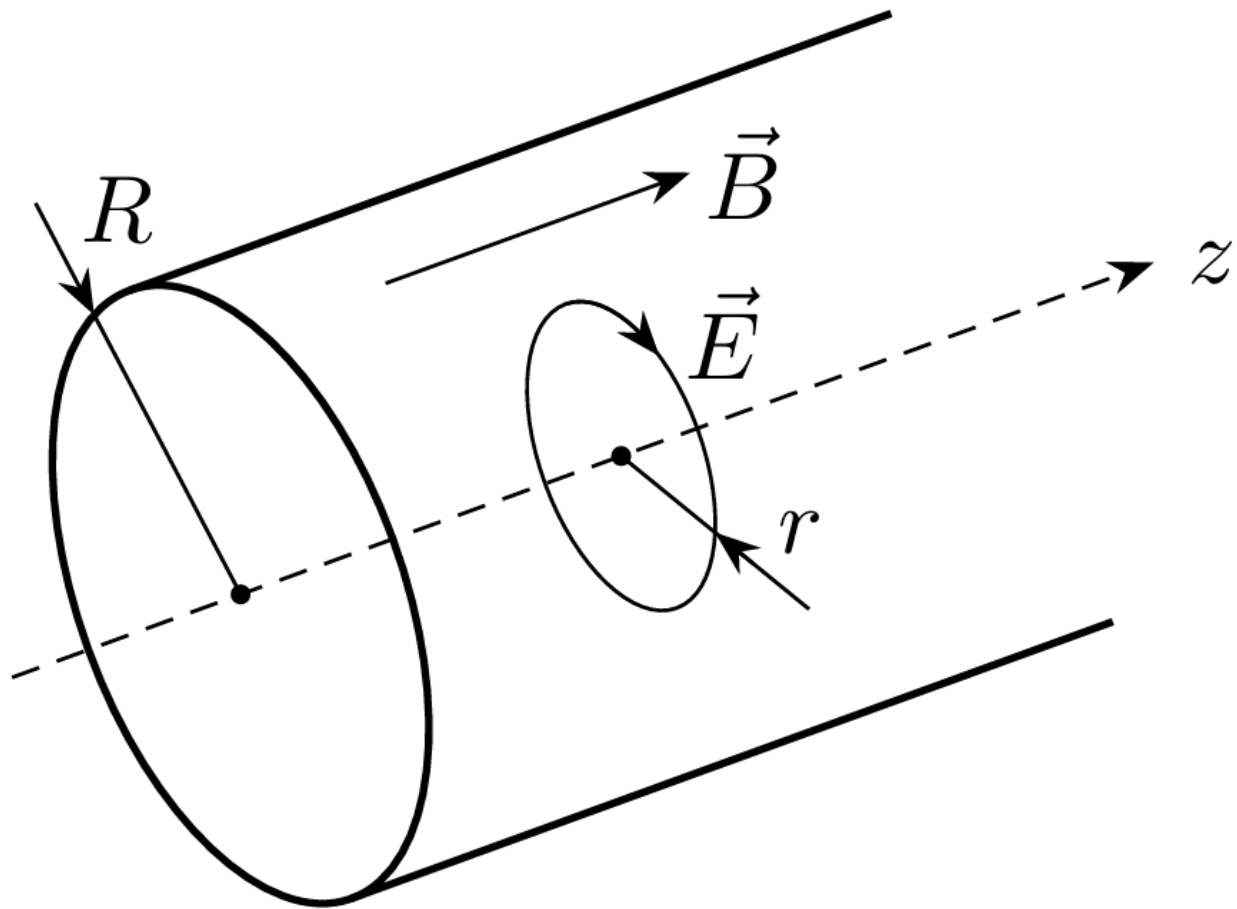


Fig. 5.

D1	Write down the expression for the tangential electric field $E(r, t)$.	0.30
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D2	Derive an expression for the average power of Joule losses P_{eddy} released in the core. 0.70 Express your answer in terms of the conductivity of the core material σ_c , its cross-sectional area S , length ℓ , and B_0 and f .	
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To reduce losses, the core is made layered along its axis - thin layers of non-conducting material are laid between the layers of the ferromagnet. In this case, Foucault currents can only be excited separately in each plate.

Single-phase transformer. Its core is assembled from plates isolated from each other to reduce losses from Foucault currents

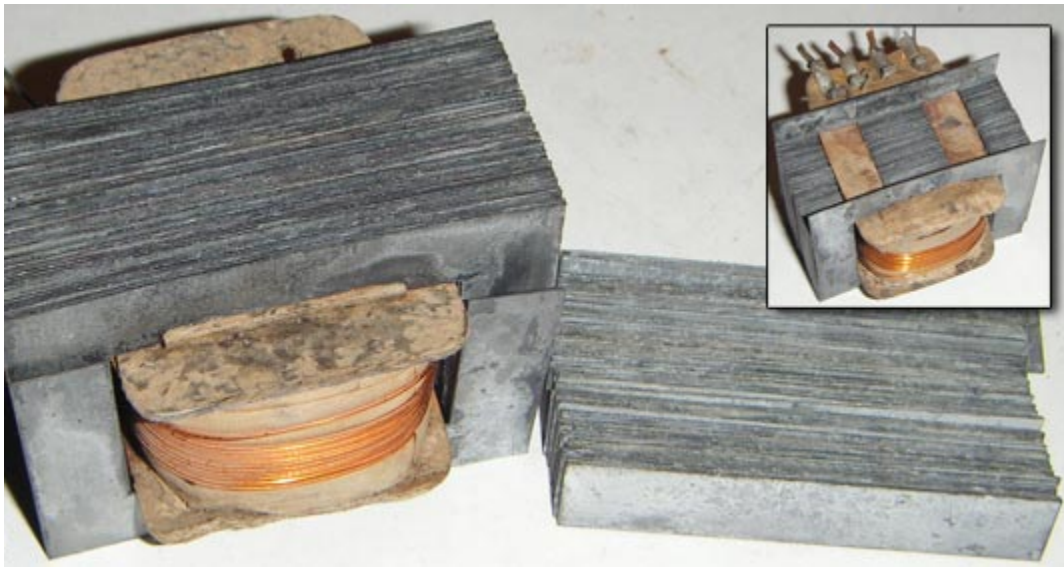


Fig. 6.

Let us consider such a plate with thickness h in an external field $\vec{B}(t) = \text{Re} [B_0 \hat{y} e^{-i\omega t}]$ directed parallel to it.

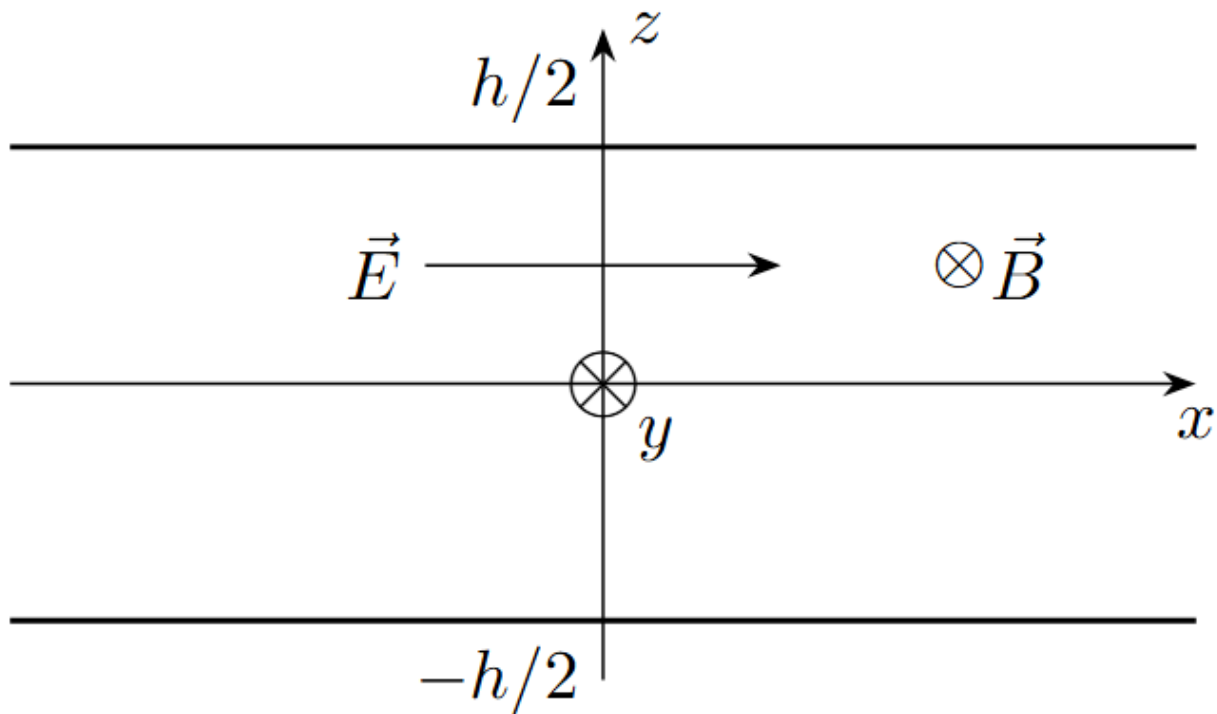


Fig. 7.

It is easy to show that in such a system the excited electric field is also parallel to the plate and perpendicular to the magnetic field. We will look for it in the form:

$$\vec{E}(z, t) = \text{Re} [A z \hat{x} e^{-i\omega t}].$$

D3 Find the factor A .

0.30

- D4** Write down an expression for the power dP/dV released in such a plate per unit volume, averaged over the entire height of the plate, as well as over time. The response may include S , ℓ , B_0 , h , σ_c and f . Let now the core be assembled from plates with the parameters indicated above in the problem. The load R is still connected to the secondary winding. Express the power P'_{eddy} that would be released in the core due to Foucault currents in terms of the parameters given in the problem. Calculate this power. **1.40**

Part E. Hysteresis (1.2 points)

Finally, let's look at the effect of hysteresis. We assume that under the influence of the magnetic field strength created by the current in the windings, the magnetic field induction inside the core describes the magnetization curve shown in the figure below. The width of this curve along H is constant, the magnetic field varies in the range from $-B_s$ to $+B_s$ (the so-called saturation induction) and is zeroed at $\pm H_c$ (the so-called coercive force).

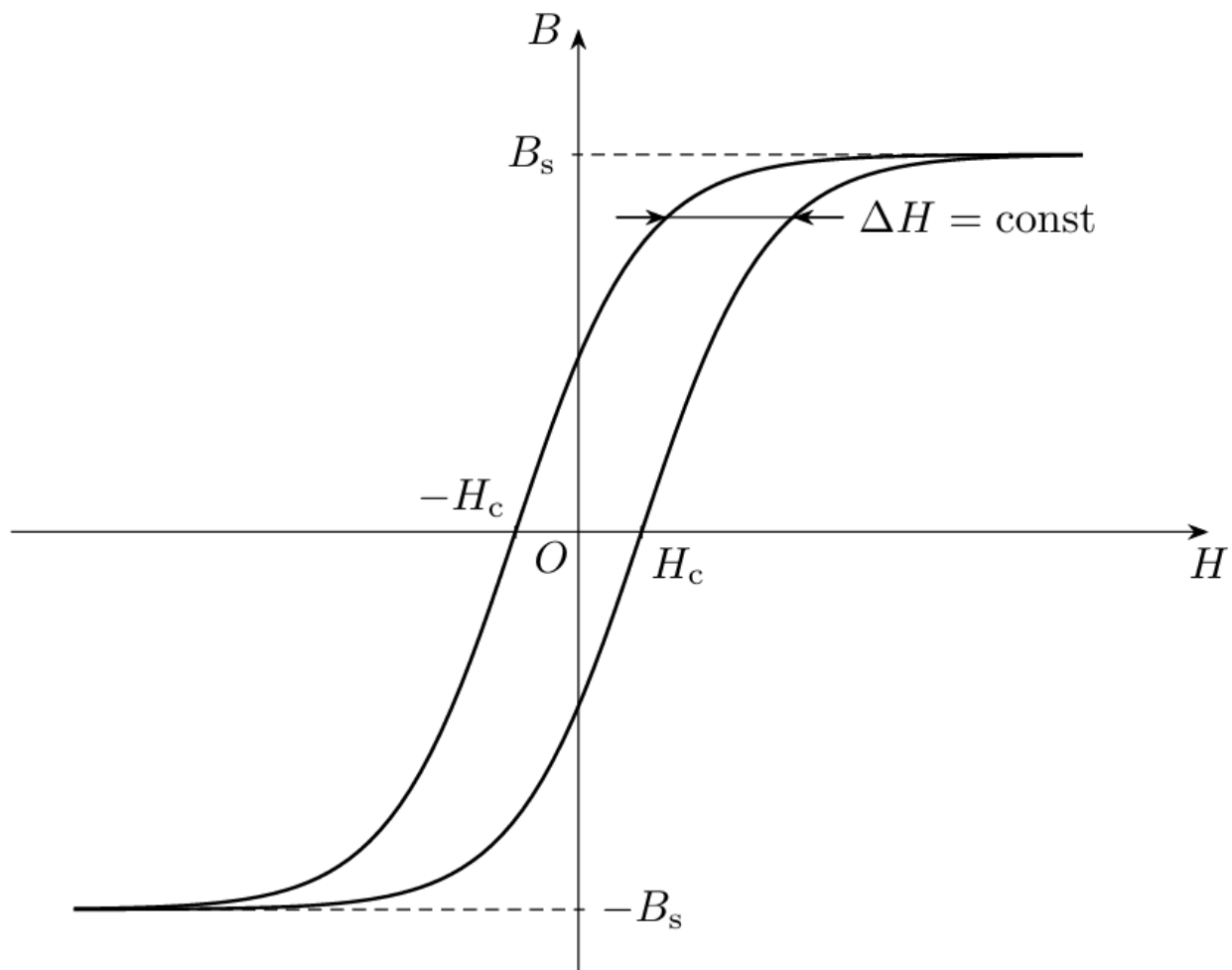


Fig. 8.

- E1** For simplicity, consider a transformer without load. Record the energy that the source imparts to the core when current I flows through the primary winding and the flux through the core changes to $d\Phi$. Then express this energy in terms of H , dB and core parameters. **0.40**

E2 We will assume that the last expression is valid in an arbitrary case. Write down the expression for the average power P_{hyst} hysteresis losses. Express your answer in terms of H_c , B_s and the quantities given at the beginning of part C. Also get the numerical value. **0.30**

E3 Assuming that all contributions calculated in parts C, D and E are small and independent, calculate the final efficiency of the transformer η . **0.50**

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